Bantay-Baha: Team Explainer

Acronyms, models, how the pipeline works in plain terms, and a simplified flowchart. For G is Cash, PJDSC 2026. Companion to the concept proposal.

1. The pipeline in plain terms

The one-line version

A typhoon track goes in. Out comes a ranked list of provinces, each with a forecast of cases for the next few weeks (with a range), the share of people who will not be able to reach a hospital, and a suggested action. Health officers use it to decide where to send supplies and teams before the roads close.

The analogy

Think of a weather forecaster who also knows the traffic. The forecaster says "80% chance of a dengue surge in Albay next month". The traffic map says "and 30% of Albay cannot get to a hospital because the bridge and the ferry are out". Multiply the two and you get where help matters most. Bantay-Baha is that forecaster and that traffic map in one tool.

Step by step

Step 0. Build one big table We take the DOH dengue file (cases per province per week, 2016 to early 2021) and line it up with everything else per province per week: rainfall from NASA satellites, how much of the province was flooded (from radar satellites), how flood-prone it is (from UP NOAH maps), and whether a typhoon passed nearby that week (from the IBTrACS track archive). Every row is one province in one week. This is the table every model reads.

Step 1. Where did it flood? (Layer 1) For each typhoon we compare a radar satellite image from before the storm with one from after. Land that suddenly looks like water is flood. We count how much of each province, and how many people, were under water. We also add up rainfall along rivers, so a town at a river mouth "feels" the rain that fell upstream. Together these numbers say how exposed each province was to that storm.

Step 2. Who gets sick next? (Layer 2) Dengue does not rise everywhere at once. Some provinces rise first, others follow a few weeks later. We let an algorithm (PCMCI+) look at years of weekly case curves, remove the shared rainy-season rhythm so it is not fooled by "everyone rises in August", and find which provinces reliably lead which. The result is a map of arrows between provinces. Provinces with many arrows pointing out of them are sentinels: when they rise, watch the ones they point to. Provinces that move together form groups that should be handled together.

Step 3. How many cases next month? (the forecast) A tree-based model (XGBoost) learns from the big table: given a province's recent cases, the week of the year, recent rain, flood exposure, and what its sentinel provinces did, how many cases come next? It predicts a range, not a single number, because a range is what a decision needs ("most likely 120, but could reach 300"). We train only on earlier years and test on later ones, exactly as if we had been forecasting in real time, and we grade it the way international forecast hubs grade flu forecasts. We also run two pre-trained "foundation" forecasters as a sanity check: if a model that was never trained on our data does as well as ours, we say so.

Step 4. Who can't reach a hospital? (Layer 3) We build a map of roads and vehicle ferries with every hospital marked. For every town we compute how long it takes to reach the nearest hospital on a normal day. Then we "run the storm": ferries inside the typhoon's wind field are cancelled, roads inside the flooded area are slowed or closed, and we recompute. Towns that now have no route, or whose trip became unreasonably long, are cut off. We do this two ways (on the road network and on a grid of travel

times) because the road map is incomplete in places; when the two disagree, we report the disagreement instead of hiding it.

Step 5. Combine and rank For each province: Risk = the probability that cases in the next four weeks exceed what that province normally sees in a bad month. Access loss = the share of its people cut off from a hospital. Priority = Risk x Access loss. A province with a high surge risk but all roads open scores lower than one with the same risk and half its people stranded. The dashboard shows the ranked list, a map, the reasons (which inputs drove the risk), and an action tier: monitor, pre-position supplies, or send mobile clinics before the roads close.

Step 6. Check ourselves We replay four real typhoons (Ompong 2018, Tisoy 2019, Rolly 2020, Ulysses 2020) and ask: did the provinces we ranked highest actually see more cases in the weeks after? We also test how much the ranking changes if we assume slower roads or a bigger storm buffer, and we ask three real health or disaster officers to use the tool and score it.

What to say to a judge in one breath

- Prior work forecasts dengue from climate, or maps flood access loss, or finds outbreak source regions. Nobody has put the three together for a specific storm.
- We use observed flood extent from radar satellites, not just rainfall.
- Our lead-lag graph is causal-style (PCMCI+), not plain correlation, which answers the "correlation is not causation" question before it is asked.
- Our forecasts come with honest ranges, graded like CDC and ECDC forecast hubs grade theirs, and benchmarked against 2025-2026 foundation models.
- Ferry links come from a real MARINA inventory, which matters in an archipelago.
- Leptospirosis has no forecasting model in the Philippines at all; digitising the DOH weekly reports to build one is our stretch goal.

2. Simplified flowchart

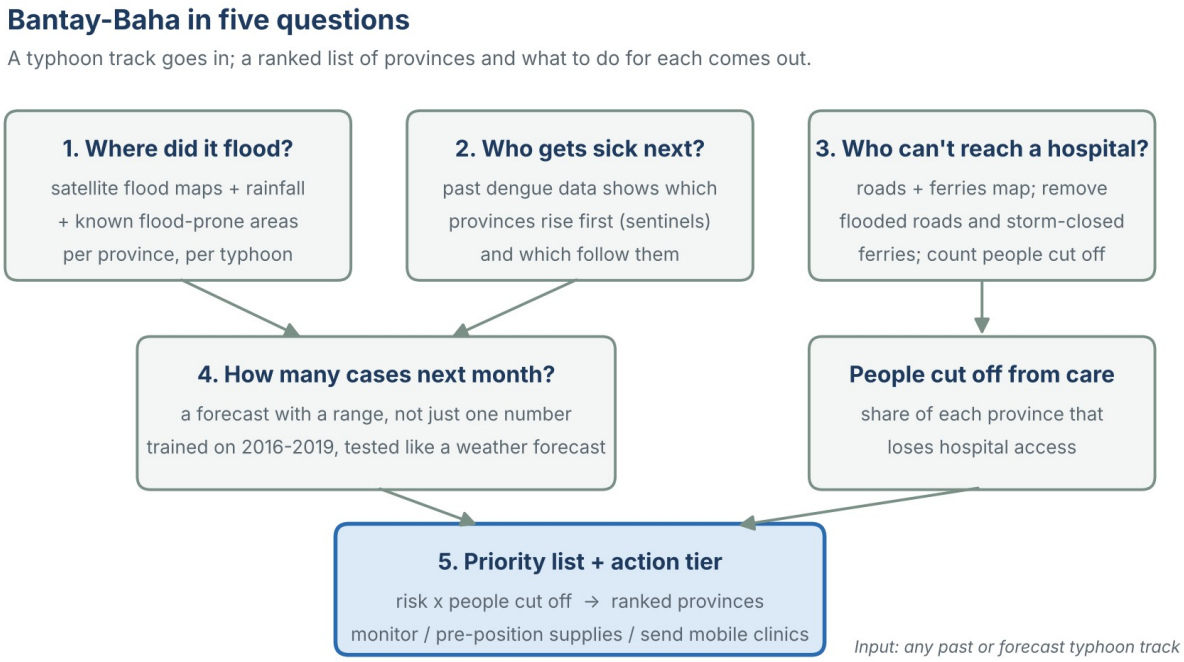


Figure A. Bantay-Baha in five questions. Boxes 1 and 2 feed the forecast (4); box 3 feeds the access-loss estimate; both meet in the priority list (5).

Technical version (same pipeline, proposal labels)

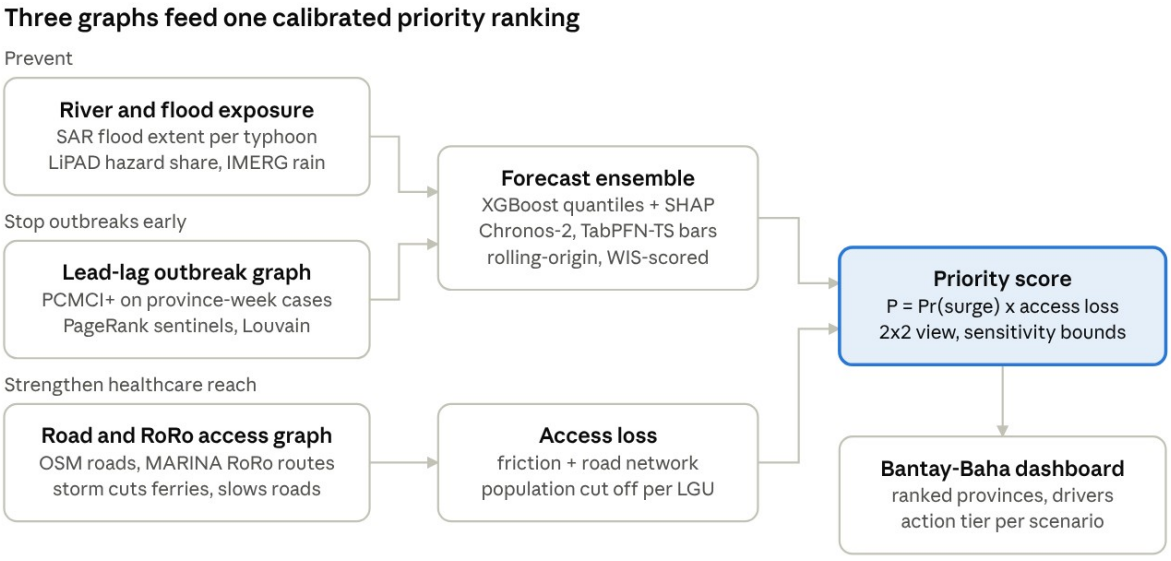


Figure B. The three-graph pipeline as drawn in the concept proposal.

3. Models and methods overview

Each model answers one question. The forecasters are compared on the same held-out weeks with the same score, so the table in the final deck will show who won and by how much.

Model / method	What it does (plain words)	Role in Bantay-Baha	Why this one	Library
Seasonal-naive baseline	Predicts "same as this week last year"	The bar every model must beat	Dengue models often fail to beat it; proving we do is the credibility test	pandas
XGBoost (quantile or Poisson objective)	Hundreds of small decision trees, each fixing the errors of the last; can output a range (quantiles) or a count rate (Poisson)	Core forecaster for 1-4 weeks ahead	Best-performing family on small tabular panels; trains in seconds; SHAP explains it; native uncertainty	xgboost >= 2.0
Chronos-2	A pre-trained "GPT for time series" from Amazon; forecasts any series without training, and can use extra inputs like rainfall	Zero-shot benchmark and ensemble member	Top of 2025-2026 forecasting leaderboards and epidemic benchmarks; runs on the 5090 in seconds	chronos-forecasting
TabPFN-TS	A pre-trained model for tables, applied to time series; works well on little data	Second zero-shot benchmark	Beat SARIMA/Prophet on Brazilian dengue in a 2025 study; fine-tunes in under a minute	tabpfn-time-series
MPNN-LSTM (or TFT)	Deep model: neighbours share information through the graph, then a sequence model predicts	Ablation only (we show we tried it)	Deep models rarely beat trees on 81 series x 200 weeks; reporting the comparison is more honest than hiding it	torch-geometric-temporal / pytorch-forecasting
PCMCI+	Finds which provinces' case curves lead which, after removing shared seasonality and other provinces' influence	Builds the lead-lag outbreak graph	Plain correlation confuses "both rise in August" with "A leads B"; PCMCI+ controls for that	tigramite
PageRank	Scores nodes by how many important nodes point at them	Picks sentinel provinces (the ones that rise first)	Simple, well-known, explainable	networkx
Louvain	Splits a network into tightly-linked groups	Finds provinces that move together, for coordinated response	Fast and standard	networkx / igraph
Sentinel-1 change detection	Compares radar images before and after a storm; pixels that got much darker are water	Flood extent per typhoon	Radar sees through clouds; the UN-SPIDER recipe is standard practice	Google Earth Engine
Downstream rainfall routing	Adds up rain along the river network so downstream towns "feel" upstream rain	Extra flood feature	Cheap, physically sensible, keeps the river graph from the original proposal	networkx + HydroRIVERS
Shortest path on road + ferry network	Finds the fastest route from every town to the nearest hospital, before and during a storm	Access loss (network method)	Direct, interpretable; ferries from a real MARINA inventory	networkx / OSMnx
Friction-surface cost distance	Same question on a grid: each 1 km cell has a crossing time; flooded cells become impassable	Access loss (raster method)	Second opinion that does not depend on OSM road completeness	Google Earth Engine (Malaria Atlas Project)
SHAP	Attributes each prediction to its inputs	The "why" column in the dashboard	Officers need reasons, not just numbers	shap
EnbPI (conformal prediction)	Wraps any forecaster and widens or narrows its range until it is honest	Cross-check on the quantile intervals	Guarantees coverage without assuming a distribution	mapie
WIS / coverage scoring	Scores a forecast range: penalises wide ranges and misses	How we grade every model	Same rules as CDC and European forecast hubs	own code / scoringutils

4. Acronyms and terms

Acronym	Meaning	In plain words / where it appears
PJDSC	Philippine Junior Data Science Challenge	The competition (UP Data Science Society)
DOH	Department of Health	National health agency; publishes the surveillance data we use
DOH-Epi / EB	DOH Epidemiology Bureau	The DOH unit that runs disease surveillance
PIDSR	Philippine Integrated Disease Surveillance and Response	DOH system that collects weekly case reports (dengue, lepto, etc.)
FHSIS	Field Health Services Information System	DOH annual health statistics reports
NHFR	National Health Facility Registry	DOH list of hospitals and clinics with locations
LGU	Local Government Unit	Province, city, municipality or barangay government
DRRM / DRRMO	Disaster Risk Reduction and Management (Office)	Local disaster office; one of our target users
PSA	Philippine Statistics Authority	Census and geographic codes
PSGC	Philippine Standard Geographic Code	Official code for every region, province, city and barangay; our join key
CPH	Census of Population and Housing	The 2020 census; gives population per area
HDX	Humanitarian Data Exchange	UN open-data site hosting the DOH dengue file, boundaries and population maps
COD-AB	Common Operational Dataset, Administrative Boundaries	Official boundary shapefiles for the Philippines on HDX
PAGASA	Philippine Atmospheric, Geophysical and Astronomical Services Administration	National weather agency
IBTrACS	International Best Track Archive for Climate Stewardship	NOAA archive of typhoon tracks (position, wind, time)
GPM IMERG	Global Precipitation Measurement, Integrated Multi-satellitE Retrievals	NASA satellite rainfall, 10 km grid, daily
CHIRPS	Climate Hazards Group InfraRed Precipitation with Station data	Another satellite-plus-station rainfall dataset, 5 km
SAR	Synthetic Aperture Radar	Radar satellite imaging that sees through clouds; Sentinel-1 is the satellite
GEE	Google Earth Engine	Free cloud platform for processing satellite images
CEMS / EMSR	Copernicus Emergency Management Service / Rapid mapping activation	EU flood maps made during disasters (EMSR312 = Ompong, EMSR478 = Rolly)
UN-SPIDER	UN Platform for Space-based Information for Disaster Management	Publishes the standard recipe we follow for SAR flood mapping
GFD	Global Flood Database	Satellite record of 913 past floods, 2000-2018
LiPAD	LiDAR Portal for Archiving and Distribution	UP DOST portal hosting flood hazard maps
UP NOAH	Nationwide Operational Assessment of Hazards	UP hazard-map programme (flood, landslide, storm surge)
DEM / FABDEM / HAND	Digital Elevation Model / Forest-and-Buildings-removed DEM / Height Above Nearest Drainage	Terrain height data and a flood-susceptibility measure derived from it
HydroRIVERS	(dataset name)	Global river network where every river segment points to the one downstream
OSM	OpenStreetMap	Free world map; source of roads and ferry lines
OSMnx	(Python library)	Downloads OSM roads into a network for routing
RoRo	Roll-on/Roll-off	Vehicle ferries linking islands
MARINA	Maritime Industry Authority	Agency whose RoRo route inventory we use (via the roroph package)

Acronym	Meaning	In plain words / where it appears
MAP	Malaria Atlas Project	Publishes the global "friction surface" (how long it takes to cross each 1 km cell)
2SFCA	Two-Step Floating Catchment Area	Standard method for measuring healthcare access (supply vs demand within travel time)
WorldPop / HRSL	WorldPop / High Resolution Settlement Layer	Gridded population maps (100 m / 30 m)
RWI	Relative Wealth Index	Meta's poverty map; optional vulnerability weight
CCHAIN	Climate Change and Health Analytics Information Network (Project CCHAIN)	Barangay-level health + climate dataset for 12 cities
PCMCI+	Peter and Clark Momentary Conditional Independence (plus)	Algorithm that finds which time series lead which, controlling for the others
PageRank	(Google's ranking algorithm)	Scores how "influential" a node is in a network; we use it to find sentinel provinces
Louvain	(community detection algorithm)	Groups nodes that are tightly connected; we use it to find provinces that move together
XGBoost / LightGBM / GBM	eXtreme Gradient Boosting / Light GBM / Gradient Boosting Machine	Tree-based machine-learning models; our core forecaster
GNN	Graph Neural Network	Deep-learning model that works on networks; our ablation
MPNN-LSTM	Message Passing Neural Network + Long Short-Term Memory	A GNN that mixes neighbour information with a time model
TFT	Temporal Fusion Transformer	Deep forecasting model with attention; alternative ablation
TS-FM	Time-Series Foundation Model	Large pre-trained forecaster used zero-shot (Chronos-2, TimesFM, TabPFN-TS)
SHAP	SHapley Additive exPlanations	Explains which inputs pushed a prediction up or down
EnbPI	Ensemble Batch Prediction Intervals	Conformal method that gives honest prediction ranges for time series
WIS	Weighted Interval Score	Standard score for probabilistic epidemic forecasts (lower is better)
MAE / MASE	Mean Absolute Error / Mean Absolute Scaled Error	Point-forecast error measures
CRPS	Continuous Ranked Probability Score	Like WIS but for a full distribution
CDC FluSight	US Centers for Disease Control flu forecasting hub	Where the 23-quantile and WIS conventions come from
DLNM	Distributed Lag Non-linear Model	Statistical model for delayed effects (rain today, cases two weeks later)
NB / Poisson	Negative Binomial / Poisson distribution	Probability models for counts of cases
SUS	System Usability Scale	10-question usability questionnaire (score 0-100)
FOI / eFOI	Freedom of Information	Legal request route for government data not published online
API	Application Programming Interface	A way for programs to fetch data from a service